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Learning GitHub Actions

Automation and Integration of CI/CD with GitHub

Brent Laster

Foreword by Julian C. Dunn

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Learning GitHub Actions

Maximize your software development productivity with GitHub Actions, the continuous integration and automation platform that works seamlessly with GitHub. With this practical book, author Brent Laster shows software developers, SREs, and DevOps engineers how to gain the maximum benefit from Actions. You'll learn how to leverage this feature in your daily work with GitHub to achieve greater simplification, standardization, and automation.

This book explains the platform, components, use cases, implementation, and integration points of actions, so you can leverage them to provide the functionality and features that today's complex pipelines and software development processes need. You'll learn how to design and implement automated workflows that respond to common events like pushes, pull requests, and review updates—and how to use Actions to implement functionality from simple validation to complex pipelines.

You'll learn how to:

- Understand the structure, syntax, and semantics of GitHub Actions
- Incorporate actions into your automation and processes
- Create custom actions with Docker, JavaScript, or shell approaches
- Troubleshoot and debug workflows that use actions
- Combine actions with GitHub APIs and other integration options
- Securely implement workflows with GitHub Actions
- Migrate from using other CI/CD platforms to GitHub Actions

"Whether you are new to CI/CD and starting with GitHub Actions as your first product in this space or are already a CI/CD expert and migrating from another tool, Brent's book has the right balance of information to help you become productive quickly."

—**Julian C. Dunn**
Senior Director of Project Management, GitHub Actions

"If you're looking to master automation in software development, *Learning GitHub Actions* is a comprehensive guide you can't miss."

—**Taylor Dolezal**
Head of Ecosystem, CNCF

Brent Laster is an R&D DevOps director at SAS. He is a global trainer, author, and speaker on open source technologies.

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US $65.99 CAN $82.99
ISBN: 978-1-098-13107-4

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Learning GitHub Actions
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Printed in the United States of America.

Published by O'Reilly Media, Inc., 1005 Gravenstein Highway North, Sebastopol, CA 95472.

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August 2023: First Edition

Release History for the First Edition
2023-08-16: First Release

See *https://oreilly.com/catalog/errata.csp?isbn=9781098131074* for release details.

978-1-098-13107-4

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To all my family and friends who have helped me write the best chapters of my life.

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Foreword

The fundamental concepts of continuous integration/continuous delivery (CI/CD) have now been around for several decades, since Martin Fowler and Matthew Foemmel of Thoughtworks first popularized CI in their seminal essay of September 2000, and Jez Humble and Dave Farley wrote about CD in their 2010 book *Continuous Delivery: Reliable Software Releases Through Build, Test, and Deployment Automation* (Addison-Wesley Professional). Yet it has taken years for widespread adoption of tools for CI/CD and for the notion of a software delivery pipeline to take root. I believe this is because three fundamental socio-technical changes in how we do software development had to occur first:

- Development had to become collaborative, rather than performed by isolated, individual engineers. This was driven first by a truly distributed version control system (Git) and then accelerated through pull-request platforms like GitHub.
- Widespread adoption of agile practices needed to occur. Motivated by metrics from DevOps leaders like DevOps Research Associates (DORA) that, contrary to intuition, showed more frequent delivery of smaller changes reduces risk, savvy engineering leaders drove the implementation of frequently used build pipelines where software was continuously built, tested, and deployed directly to customers—sometimes dozens of times per day.
- The overburdening of traditional IT operations functions with increased complexity drove massive cultural changes via the DevOps movement. This led to a you-build-it, you-own-it approach to software operations where, increasingly, developers take full ownership for the success, failure, and performance of their software in production, rather than throwing code over the wall to a release engineering team that would operate a build process and somehow "add quality" to software that didn't have it already.

All these changes mean that GitHub Actions, as a relatively recent entrant to the CI/CD pipeline and automation category, is a substantially different product from

incumbents. It is natively integrated into GitHub, making it a natural fit for developers who are already familiar with storing their source code there. GitHub Actions is also designed around the concept of a workflow, which can be used to create CI/CD pipelines but also handle any other kind of software automation tasks like managing open issues and tasks that open source and enterprise developers alike need to perform in the course of their work. Finally, GitHub Actions, as the name would suggest, is based on the notion of an action: a reusable component that helps to encapsulate common tasks and reduce repetition when authoring workflows. The GitHub Marketplace offers nearly 20,000 actions at the time of writing, making it easy for developers, DevOps engineers, and site reliability engineers to get started with any kind of build automation task.

Although GitHub Actions is a sophisticated product, learning it doesn't need to be complicated. Brent Laster has written an excellent book that relies on progressive disclosure, starting with the most basic concepts to get you up and running with GitHub Actions while also providing a comprehensive tour of GitHub Actions' most advanced features to help you to optimize your use of the product as you adopt it across your organization. My team and I have been delighted to partner with Brent in ensuring that the content covered here is as current as possible. Whether you are new to CI/CD and starting with GitHub Actions as your first product in this space or are already a CI/CD expert and migrating from another tool, Brent's book has the right balance of information to help you become productive quickly.

We wish you the greatest success in automating your software delivery processes.

— Julian C. Dunn
Senior Director of Product Management
GitHub Actions

Preface

> *Releasing software should be easy.… Automate almost everything, and keep everything you need to build, deploy, test, and release your application in version control.*
>
> —David Farley, *Continuous Delivery: Reliable Software Releases Through Build, Test, and Deployment Automation*

Back in 1968, the London Underground in the United Kingdom needed a digital sign to warn passengers to be careful while crossing the gaps between train doors and station platforms. Since data storage for such signs was very expensive back in the day, they chose a very short phrase to help keep riders alert: "mind the gap."

These days, the word "mind" is less commonly used, but the intent to bring awareness to missing parts or things that can trip you up and to act on them is still meaningful. And it is just as important when we apply the idea to business and technical processes that can benefit from automation.

From its inception in 2008, GitHub has filled gaps in terms of allowing users to collaborate and build communities around open source software. And it has done this very well. It is challenging not to overestimate the significance of the SaaS hosting model that GitHub pioneered and the collaborative ecosystem it has built around it. Yet up until a few years ago, there was one key piece of that ecosystem that was clearly missing—a tightly integrated automation platform for key functions like CI/CD.

Certainly, there has been no shortage of applications that have worked to fill that gap. Tools such as Jenkins, Travis CI, CircleCI, Azure DevOps, and more have provided integration methods through various approaches such as webhooks. However, users of GitHub still had to go *outside* their collaboration environment to use another application to get the basic functionality they needed. All of that has changed with the addition of GitHub Actions.

Actions is challenging to classify with a standalone designation. It is a logical extension of the larger GitHub model. And while this is not a general book on GitHub, I have tried to write it in such a way that you can see how GitHub Actions plays with